HOT SPOTS AND EXTREMUM LOCALIZATION ON THE REGULAR DODECAHEDRON

Abstract. We prove that every eigenfunction for the first positive Neumann eigenvalue of a solid regular dodecahedron attains its global maximum and minimum only on the boundary. The eigenspace is three-dimensional and is canonically isomorphic to the ambient reflection representation through the integral of the gradient. An invariant cone for the Jacobian of the coordinate Neumann heat flow yields a common strictly increasing direction for every dominant eigenfunction, including at intersections of internal symmetry mirrors. We give all 28 generators and all projection identities of the exact algebraic certificate. We then locate extrema in explicit unions of boundary flag triangles according to the eigenfunction parameter. On one parameter mirror the maxima are confined to a physical edge. For a parameter pointing toward a vertex, the unique maximum is that vertex and the unique minimum is its antipode. The boundary arguments include weak tangential differentiation at physical edges and a local Neumann positivity lemma valid at vertices. TikZ diagrams distinguish exact extremal points from the containing regions proved for general eigenfunctions.

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1. Introduction and main results

Let $P \subset \mathbb{R}^3$ be the interior of a solid regular dodecahedron. Its first positive Neumann eigenvalue is

$$\mu_1(P) = \inf_{\substack{u \in H^1(P)\setminus\{0\} \\ \int_P u = 0}} \frac{\int_P |\nabla u|^2 \, \mathrm{d}x}{\int_P u^2 \, \mathrm{d}x},$$

and $E_1(P)$ denotes the corresponding real eigenspace. The Neumann equation is understood weakly:

$$\int_P \nabla u \cdot \nabla \eta \, \mathrm{d}x = \mu_1(P) \int_P u\eta \, \mathrm{d}x \qquad (\eta \in H^1(P)).$$

Convex-domain regularity gives $u \in H^2(P)$ and hence a continuous representative on $\overline{P}$. The strict hot spots assertion is

$$\min_{\overline{P}} u < u(x) < \max_{\overline{P}} u \qquad (x \in P, 0 \neq u \in E_1(P)).$$

The higher-dimensional variational work of Kennedy and Rohleder [4] establishes hot spots and directional monotonicity for lip domains and certain symmetry sectors. Reflection symmetries also organize the first eigenspace on a regular polyhedron. To obtain the derivative signs needed on the dodecahedron, we evolve the coordinate map by the Neumann heat equation and work with its full derivative matrix. The resulting chamber problem has four orthogonal boundary projections on the nine-dimensional space of real matrices. We give an explicit invariant convex cone for these projections and pass its inequalities to the first eigenspace. The proof uses the ordinary Neumann problem on the three-dimensional solid, with no smoothing of its edges or vertices.

1.1. Geometry and canonical parameters.

All assertions are invariant under translations, orthogonal maps, and dilations. We use the following normalization:

$$\varphi = \frac{1+\sqrt{5}}{2}, \qquad \alpha_1 = (1,0,0), \quad \alpha_2 = \tfrac{1}{2}(-\varphi, \varphi^{-1}, -1), \quad \alpha_3 = (0,0,1), \tag{1.1}$$

$$b_1 = (1, \varphi^2, 0), \qquad b_2 = (0, 2\varphi, 0), \qquad b_3 = (0, \varphi, 1). \tag{1.2}$$

Thus $|\alpha_i| = 1$ and $\alpha_i \cdot b_j = \delta_{ij}$. Set $s_i = I - 2\alpha_i \alpha_i^T$ and $W = \langle s_1, s_2, s_3 \rangle$, the group of type H_3. Its fundamental chamber is

$$C = \{x : \alpha_i \cdot x > 0 \ (1 \le i \le 3)\} = \{(x,y,z) : x, z > 0, \ y > \varphi^2 x + \varphi z\}. \tag{1.3}$$

The normalized solid and its chamber are

$$P = \{x : (wb_3) \cdot x < 1 \text{ for every } w \in W\}, \tag{1.4}$$

$$T = P \cap C = C \cap \{b_3 \cdot x < 1\}. \tag{1.5}$$

Write

$$v = \frac{b_1}{2\varphi + 1}, \qquad e = \frac{b_2}{2\varphi^2}, \qquad c = \frac{b_3}{\varphi + 2}, \qquad \Delta = \operatorname{conv}\{c, e, v\}. \tag{1.6}$$

Here v is a vertex, e an incident edge midpoint, and c an incident face center. The full pentagonal face and full physical edge containing this flag are

$$\Gamma = \overline{P} \cap \{b_3 \cdot x = 1\}, \qquad E = [s_1 v, v], \qquad \Delta = \Gamma \cap \overline{C}. \tag{1.7}$$

The geometry is proved in Section 3.

Theorem 1.1 (Eigenspace and strict hot spots)**.** *For a solid regular dodecahedron,*

$$\dim E_1(P) = 3.$$

Every nonzero $u \in E_1(P)$ attains its global maximum and global minimum on $\overline{P}$ only on ∂P. Moreover,

$$J : E_1(P) \longrightarrow \mathbb{R}^3, \qquad Ju = \int_P \nabla u \, \mathrm{d}x \tag{1.8}$$

is an isomorphism.

Define $f_a = J^{-1}a$ and $F = (f_{e_1}, f_{e_2}, f_{e_3})^T$. Then

$$E_1(P) = \{f_a = a \cdot F : a \in \mathbb{R}^3\}, \qquad F(wx) = wF(x), \qquad \int_P DF \, \mathrm{d}x = I.$$

In particular, this is a description of the whole eigenspace, including every linear combination of the three canonical modes.

Theorem 1.2 (Strict ascent in a closed chamber)**.** *For every $a \in \overline{C} \setminus \{0\}$,*

$$D_{b_3} f_a(q) > 0 \qquad (q \in P \cap \overline{C}). \tag{1.9}$$

This includes internal mirror planes, their intersection axes, and the center, provided the point is interior to the solid.

For extrema on the physical boundary, write

$$\mathcal{M}(a) = \mathrm{Argmax}_{\overline{P}} f_a, \qquad \Updownarrow(a) = \mathrm{Argmin}_{\overline{P}} f_a, \qquad W_a = \{w \in W : wa = a\}.$$

Every parameter can be moved into $\overline{C}$ by a symmetry.

Theorem 1.3 (Boundary regions for all parameters)**.** *Let $a \in \overline{C} \setminus \{0\}$. Then*

$$\mathcal{M}(a) \subset \bigcup_{w \in W_a} w\Delta, \qquad \Updownarrow(a) = -\mathcal{M}(a). \tag{1.10}$$

If $a \in C$, every maximum lies in the single flag triangle Δ. If a is a positive multiple of b_3, every maximum lies in the single pentagonal face Γ.

Theorem 1.4 (Edge localization and exact vertex extrema)**.** *Suppose $a = t_1 b_1 + t_2 b_2$, where $t_1, t_2 \geq 0$ and $a \neq 0$. If $t_1 > 0$, then*

$$\mathcal{M}(a) \subset (e, v].$$

If $t_1 = 0$, then $\mathcal{M}(a) \subset E$. For every $w \in W$ and $\lambda > 0$, one has the exact equalities

$$\mathcal{M}(\lambda w b_1) = \{wv\}, \qquad \Updownarrow(\lambda w b_1) = \{-wv\}. \tag{1.11}$$

All minimum-set containments are obtained from the maximum-set containments by central inversion.

The location statements for general modes are proved containments. The exact vertex locations in (1.11) are proved equalities. The precise maximizing point on the containing edge for f_{b_2}, and the precise maximizing points inside the containing face regions for general parameters, are not determined by these theorems. In the figures, red denotes maxima and blue denotes minima; captions specify when a colored set is a containing region rather than the full extremal set.

1.2. **Outline of the proof.** Section 2 proves a flat Hessian identity and injectivity of the mean-gradient map. Section 3 identifies the reflection geometry, the entire first eigenspace, and the signs of the root modes. Section 4 gives the exact matrix cone, including every generator and all projection identities. The coordinate heat flow, its derivative boundary problem, cone preservation, and the positive first spectral limit are established in Section 5. Stabilizer averaging makes the derivative inequalities strict on all internal mirror strata in Section 6, where reflection comparisons complete the hot spots proof. Section 7 first establishes strict Neumann positivity at physical edges and vertices, then proves the general boundary regions. Tangential inequalities on an open face and a weak derivative argument along an actual physical edge give the exact vertex extrema and the additional edge localization. The final illustrations record the proved locations and their scope.

2. The first eigenspace on a convex polyhedron

In this section $P \subset \mathbb{R}^3$ is the interior of any bounded full-dimensional convex polyhedron and $\mu = \mu_1(P)$. The homogeneous Neumann operator domain is contained in $H^2(P)$; this follows from [1, Proposition 4.8], since bounded convex domains are quasi-convex in the terminology of that paper. In dimension three, Sobolev embedding on a Lipschitz domain gives continuous representatives on $\overline{P}$. Interior eigenfunctions are real analytic. On a relatively open flat facet, the weak Neumann equation can be extended evenly to give the usual smooth local reflection across that facet.

Lemma 2.1 (Density with linear face conditions)**.** *Let V be a finite-dimensional Euclidean space, and prescribe a linear subspace $S_H \subset V$ on each face H of P. In the space*

$$\{X \in H^1(P;V) : \operatorname{Tr}_H X \in S_H \text{ on every open face } H\},$$

restrictions of functions smooth on a neighborhood of $\overline{P}$, vanishing on a neighborhood of all edges and vertices and satisfying the same face conditions, are dense in H^1.

Proof. Let S be the union of edges and vertices and $d(x) = \operatorname{dist}(x, S)$. Since S is a finite union of segments and points, its tubular neighborhood has volume $m(r) := |\{d < r\} \cap P| \le Cr^2$ for sufficiently small r. For sufficiently small $0 < \varepsilon < 1/2$, put

$$\chi_\varepsilon(x) = \begin{cases} 0, & d(x) \le \varepsilon^2, \\ \dfrac{\log(d(x)/\varepsilon^2)}{|\log \varepsilon|}, & \varepsilon^2 < d(x) < \varepsilon, \\ 1, & d(x) \ge \varepsilon. \end{cases}$$

The distance function is 1-Lipschitz. Stieltjes integration by parts, using $m(r) \le Cr^2$ and discarding the nonpositive lower endpoint term, therefore gives

$$\int_P |\nabla \chi_\varepsilon|^2 \, \mathrm{d}x \le \frac{1}{|\log \varepsilon|^2} \int_{\varepsilon^2}^{\varepsilon} r^{-2} \, dm(r) \le \frac{C(1 + 2|\log \varepsilon|)}{|\log \varepsilon|^2} \longrightarrow 0.$$

First replace X by the radial truncation $X_M = X \min\{1, M/|X|\}$, with value zero where $X = 0$. The truncation is 1-Lipschitz, preserves every linear subspace S_H, and hence preserves the face conditions by the Sobolev trace and chain rules. It equals the identity on $\{|X| < M\}$, while $|DX_M - DX| \le 2|DX|$ almost everywhere. Dominated convergence thus gives $X_M \to X$ in H^1 as $M \to \infty$. For fixed M, the product rule and

$$\|X_M \nabla \chi_\varepsilon\|_2 \le M \|\nabla \chi_\varepsilon\|_2 \longrightarrow 0$$

give $\chi_\varepsilon X_M \to X_M$ in H^1 as $\varepsilon \downarrow 0$; all other terms converge by dominated convergence. Multiplication by a scalar preserves the face conditions, and the resulting field vanishes near S. A partition of unity on its support now gives interior patches and patches meeting just one face, all a positive distance from S. If R_H is orthogonal projection onto S_H, extend the values across its geometric face plane by the involution $2R_H - I$. The S_H part extends evenly and its orthogonal complement

extends oddly; the latter has zero trace. This is an H^1 extension. Choose each face patch to be a ball centered on that face and localize compactly inside the ball before extending. The reflected field is then extended by zero outside this symmetric ball. If r_H denotes the geometric reflection, this extension obeys $\widetilde{X}(r_H x) = (2R_H - I)\widetilde{X}(x)$. Convolution with a radial mollifier preserves that identity, so on the face the mollified values belong to S_H. For sufficiently small mollifier radii the supports stay away from S and from every other face. The interior patches are mollified in the usual way. These convolutions converge in H^1 on their respective patches. Summing them and taking a diagonal sequence in the mollifier radius, ε, and M proves the claim. □

Lemma 2.2 (Flat Hessian identity)**.** *If $u \in H^2(P)$ satisfies homogeneous Neumann conditions, then*

$$\int_P |D^2 u|^2 \,\mathrm{d}x = \int_P (\Delta u)^2 \,\mathrm{d}x. \tag{2.1}$$

In particular, if $u \in E_1(P)$, both sides are $\mu \int_P |\nabla u|^2 \,\mathrm{d}x$.

Proof. First let X be smooth, tangent to every face, and zero near the edges and vertices. For $Z = X \operatorname{div} X - (X \cdot \nabla)X$, direct differentiation gives

$$\operatorname{div} Z = (\operatorname{div} X)^2 - \sum_{i,j} \partial_i X_j \partial_j X_i.$$

On each open face, with its constant unit normal ν,

$$Z \cdot \nu = (X \cdot \nu) \operatorname{div} X - X \cdot \nabla (X \cdot \nu) = 0.$$

Here X is tangent, so the last term differentiates the zero face trace tangentially. The divergence theorem gives $\int_P \operatorname{div} Z = 0$. Combining this with the pointwise identity

$$|DX|^2 - \sum_{i,j} \partial_i X_j \partial_j X_i = \sum_{i<j} (\partial_i X_j - \partial_j X_i)^2$$

proves

$$\int_P |DX|^2 \,\mathrm{d}x = \int_P (\operatorname{div} X)^2 \,\mathrm{d}x + \int_P \sum_{i<j} (\partial_i X_j - \partial_j X_i)^2 \,\mathrm{d}x.$$

Lemma 2.1, with $S_H = \nu_H^{\perp}$, extends this identity to every tangent field $X \in H^1(P; \mathbb{R}^3)$, since all first derivatives converge in L^2. In particular, no edge or vertex term is introduced in this limit. Now $X = \nabla u \in H^1(P; \mathbb{R}^3)$ is tangent by the Neumann trace condition, and its antisymmetric derivatives vanish because weak mixed derivatives commute. This proves (2.1). For an eigenfunction, $\Delta u = -\mu u$ almost everywhere and $\|\nabla u\|_2^2 = \mu \|u\|_2^2$, proving the final assertion. □

Lemma 2.3 (Uniqueness from a face patch)**.** *A Neumann eigenfunction whose Dirichlet trace vanishes on a nonempty open face patch vanishes throughout P.*

Proof. Choose a ball B centered on the relative interior of the patch, small enough that $P \cap B$ is one half-ball and its planar boundary lies in the patch. Zero Dirichlet trace allows the eigenfunction u to be extended by zero to a function $\widetilde{u} \in H^1(B)$. For every $\zeta \in C_c^\infty(B)$, its restriction to P, set equal to zero outside B, is an admissible weak Neumann test. Consequently, if λ is the eigenvalue,

$$\int_B \nabla \widetilde{u} \cdot \nabla \zeta = \int_{P \cap B} \nabla u \cdot \nabla \zeta = \lambda \int_{P \cap B} u \zeta = \lambda \int_B \widetilde{u} \zeta.$$

Thus $\widetilde{u}$ solves the Helmholtz equation weakly across the plane. Interior elliptic regularity makes it real analytic on B; since it is zero on the opposite open half-ball, it is zero throughout B. Analytic continuation of u through the connected set P finishes the proof. □

Proposition 2.4 (Mean-gradient injectivity)**.** *The map $J_P : E_1(P) \to \mathbb{R}^3$, $J_P u = \int_P \nabla u \,\mathrm{d}x$, is injective. If the orthogonal symmetry group of P acts irreducibly on $\mathbb{R}^3$, then J_P is an isomorphism and* $\dim E_1(P) = 3$.

Proof. Suppose $J_P u = 0$. Each $\partial_j u \in H^1(P)$ has mean zero. The three Poincare inequalities and Lemma 2.2 give

$$\mu\|\nabla u\|_2^2 \le \sum_j \|\nabla \partial_j u\|_2^2 = \|D^2 u\|_2^2 = \mu\|\nabla u\|_2^2.$$

Every scalar Poincare defect therefore vanishes. More explicitly, the quadratic form

$$Q(v) = \int_P |\nabla v|^2 \,\mathrm{d}x - \mu \int_P v^2 \,\mathrm{d}x$$

is nonnegative on the mean-zero subspace of $H^1(P)$ and satisfies $Q(\partial_j u) = 0$. For any mean-zero $\zeta \in H^1(P)$, the coefficient of t in the nonnegative polynomial $Q(\partial_j u + t\zeta)$ must vanish. Taking $\zeta = \eta - |P|^{-1}\int_P \eta$ and using $\int_P \partial_j u = 0$ gives

$$\int_P \nabla \partial_j u \cdot \nabla \eta \,\mathrm{d}x = \mu \int_P (\partial_j u)\eta \,\mathrm{d}x \qquad (\eta \in H^1(P)).$$

Thus every $\partial_j u$ belongs to $E_1(P)$, including when it is zero; its Neumann equation follows from variational equality. For each face with unit normal ν, the linear combination $\nu \cdot \nabla u$ therefore belongs to $E_1(P)$. Its trace on that face is zero by the original Neumann condition, so Lemma 2.3 makes it zero throughout P. The face normals span $\mathbb{R}^3$: a nonzero vector orthogonal to all of them would generate a line of translations preserving every defining half-space, contradicting boundedness. Hence $\nabla u = 0$, and connectedness and the mean-zero condition give $u = 0$.

For any orthogonal symmetry w of P, change of variables in the weak eigenvalue equation gives $u \circ w^{-1} \in E_1(P)$ and

$$J_P(u \circ w^{-1}) = w J_P u$$

by the chain rule and invariance of volume. Since $E_1(P) \neq \{0\}$, injectivity makes the image of J_P a nonzero invariant subspace. Irreducibility therefore makes this image all of $\mathbb{R}^3$, proving the isomorphism and the dimension assertion. □

3. Dodecahedral geometry and reflection eigenfunctions

The Gram matrix of the unit roots in (1.1) is

$$G = \begin{pmatrix} 1 & -\varphi/2 & 0 \\ -\varphi/2 & 1 & -1/2 \\ 0 & -1/2 & 1 \end{pmatrix}. \tag{3.1}$$

The associated chamber angles are $\pi/5, \pi/3, \pi/2$. Thus the corresponding Coxeter matrix has $m_{12} = 5$, $m_{23} = 3$, and $m_{13} = 2$. The finite-reflection-group construction and classification identify W as H_3, of order 120, with $\overline{C}$ as a fundamental chamber; see [3, Chapters 1–2]. In particular, every W-orbit meets $\overline{C}$ in exactly one point. For $q \in \overline{C}$ its stabilizer is

$$W_q = \langle s_i : \alpha_i \cdot q = 0 \rangle.$$

We will also use the resulting local chamber decomposition: at a point of $\overline{C}$, the images of $\overline{C}$ under that stabilizer fill a neighborhood of the point, up to restricting to a sufficiently small ball avoiding all other mirrors.

These facts also identify (1.4) concretely. The orbit of b_3 consists of the twelve vectors

$$(0, \pm\varphi, \pm 1), \quad (\pm 1, 0, \pm\varphi), \quad (\pm\varphi, \pm 1, 0), \tag{3.2}$$

with independent signs in each displayed pair. They form a regular icosahedron in its standard coordinate realization. Substitution, using $\varphi^2 = \varphi + 1$, shows that each s_i permutes this set, so the orbit Wb_3 is contained in it. Conversely, $\alpha_i \cdot b_3 = \delta_{i3}$ gives $W_{b_3} = \langle s_1, s_2 \rangle$. Two reflections meeting at angle $\pi/5$ generate a dihedral group of order ten. Hence $|Wb_3| = 120/10 = 12$, proving equality with the displayed set. Its convex hull contains the origin in its interior, and (1.4) is the interior of its polar. It is consequently a regular dodecahedron. The definition also shows that W preserves

P. For $x \in \overline{C}$, the maximum of $x \cdot wb_3$ is $x \cdot b_3$. Indeed, reflect a vector of this finite orbit with a negative simple-root coordinate across that root. Its pairing with x does not decrease, since

$$x \cdot (s_i z - z) = -2(\alpha_i \cdot z)(\alpha_i \cdot x) \geq 0 \qquad (\alpha_i \cdot z < 0).$$

Its pairing with a fixed $h \in C$ strictly increases at each step. Finiteness of the orbit therefore forces termination at its unique point in $\overline{C}$, namely b_3. This proves (1.5).

To compute the chamber vertices, write $x = \sum_i t_i b_i$, with $t_i = \alpha_i \cdot x$. Then

$$\overline{T} = \left\{ \sum_{i=1}^{3} t_i b_i : t_i \geq 0, \quad (2\varphi+1)t_1 + 2\varphi^2 t_2 + (\varphi+2)t_3 \leq 1 \right\}.$$

Its four vertices are therefore

$$0, \qquad v = \frac{b_1}{2\varphi+1}, \qquad e = \frac{b_2}{2\varphi^2}, \qquad c = \frac{b_3}{\varphi+2}. \tag{3.3}$$

Here v is a dodecahedral vertex, e an adjacent edge midpoint, and c an incident face center, as follows. Since the twelve normals have the same norm, $c = b_3/|b_3|^2$ satisfies every face inequality other than $b_3 \cdot c = 1$ strictly. Thus it lies in the relative interior of the face Γ and is its center, fixed by W_{b_3}. The subgroup $\langle s_2, s_3 \rangle$ fixes b_1 and has order six, because its two mirrors meet at angle $\pi/3$. At v precisely three face inequalities are equalities, with normals $(0, \varphi, 1)$, $(0, \varphi, -1)$, and $(\varphi, 1, 0)$; they are linearly independent, so v is a vertex with three incident faces. Its stabilizer is $\langle s_2, s_3 \rangle$, and the orbit Wv therefore consists of twenty vertices. The subgroup $\langle s_1, s_3 \rangle$ has order four, since its mirrors are orthogonal, and fixes e. The two physical faces meeting along the distinguished edge have normals $(0, \varphi, 1)$ and $(0, \varphi, -1)$. Their planes intersect in $y = \varphi^{-1}$, $z = 0$. The inequalities with normals $(\pm\varphi, 1, 0)$ restrict this intersection to $|x| \leq (2\varphi+1)^{-1}$; substituting this bound shows that all other face inequalities are satisfied, using $(2\varphi+1)^{-1} < 1$ and $\varphi/(2\varphi+1) + \varphi^{-1} = 1$. This proves $E = [s_1 v, v]$ and identifies $e = (0, \varphi^{-1}, 0)$ as its midpoint. The simplex description gives $\Gamma \cap \overline{C} = \Delta = \mathrm{conv}\{c, e, v\}$ and $\partial P \cap \overline{C} = \Delta$.

For completeness, the ten translates of Δ under W_{b_3} cover the entire face Γ. Apply the same terminating simple-reflection procedure to a point $q \in \Gamma$ with a negative simple-root coordinate. The index cannot be 3, since then $b_3 \cdot s_3 q = 1 - 2\alpha_3 \cdot q > 1$, contradicting $s_3 q \in \overline{P}$. Thus only s_1, s_2 are used; they fix b_3 and keep every iterate in Γ. The terminal point lies in Δ, proving the covering. The relative interiors of these triangles are disjoint by the fundamental-chamber property. Since c, e, v are respectively the face center, an edge midpoint, and an incident vertex, these are exactly the ten face flags.

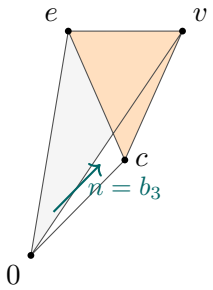


FIGURE 1. A linear projection of the flag chamber $\overline{T}$. The shaded triangle cev lies on the physical pentagonal face; the other three faces are mirrors. The arrow gives the common direction of increase in Theorem 1.2.

The reflection representation is irreducible. Indeed, the roots form a basis, so a nonzero invariant subspace contains a vector u with $u \cdot \alpha_i \neq 0$ for some i. The identity $u - s_i u = 2(u \cdot \alpha_i)\alpha_i$ puts α_i in that subspace. If $\alpha_i \cdot \alpha_j \neq 0$, applying the same identity to α_i and s_j puts α_j there as well. The

connected chain of nonzero adjacent entries in (3.1) therefore puts all three roots in the subspace, making it $\mathbb{R}^3$. Since W preserves P, Proposition 2.4 applies and gives both $\dim E_1(P) = 3$ and the isomorphism J used to define f_a.

Proposition 3.1 (Canonical covariance and reflection signs)**.** *For every orthogonal symmetry w of P and every $a \in \mathbb{R}^3$, the canonical map F satisfies*

$$f_a(wx) = f_{w^{-1}a}(x), \qquad F(wx) = wF(x), \qquad DF(wx) = wDF(x)w^{-1}. \tag{3.4}$$

The first two identities hold for $x \in \overline{P}$, and the derivative identity holds for $x \in P$. Moreover,

$$\int_P DF \, \mathrm{d}x = I, \tag{3.5}$$

and for every unit normal α to a reflection mirror of W,

$$f_\alpha(x) > 0 \quad \text{if } x \in P,\ \alpha \cdot x > 0. \tag{3.6}$$

In particular, $F(C \cap P) \subset C$.

Proof. The equivariance of J proved in Proposition 2.4 and its injectivity give $f_a \circ w = f_{w^{-1}a}$. Linearity in a then gives the vector identity in (3.4); differentiating it gives the matrix identity. Continuity extends the first two to $\overline{P}$. The (i, j) entry of $\int_P DF$ is $\int_P \partial_j f_{e_i} = \delta_{ij}$, proving (3.5).

For a unit reflection normal α, covariance makes f_α odd under s_α. The half-polyhedron $P_+ = \{x \in P : \alpha \cdot x > 0\}$ is nonempty and connected by convexity and reflection symmetry. Define the mixed form space

$$\mathcal{V}_\alpha = \{v \in H^1(P_+) : \operatorname{Tr} v = 0 \text{ on } P \cap \alpha^\perp\}.$$

There is no trace restriction on the physical boundary. For every $v \in \mathcal{V}_\alpha$, odd reflection through the mirror gives $\widetilde{v} \in H^1(P)$: the zero traces match across the mirror. The extension has mean zero and doubles both the Dirichlet integral and squared L^2 norm. Consequently, for $v \neq 0$,

$$\frac{\int_{P_+} |\nabla v|^2 \, \mathrm{d}x}{\int_{P_+} v^2 \, \mathrm{d}x} = \frac{\int_P |\nabla \widetilde{v}|^2 \, \mathrm{d}x}{\int_P \widetilde{v}^2 \, \mathrm{d}x} \geq \mu_1(P).$$

The restriction of f_α belongs to $\mathcal{V}_\alpha$ and is nonzero because $Jf_\alpha = \alpha$. Its oddness makes the energy and squared norm on the two half-polyhedra equal, so its mixed Rayleigh quotient is exactly $\mu_1(P)$. It is therefore a minimizer, as is its absolute value, since taking the absolute value preserves both integrals. The Euler–Lagrange equation gives $-\Delta|f_\alpha| = \mu_1(P)|f_\alpha|$ in P_+. Interior regularity makes this a classical equation; the nonnegative solution is superharmonic, so the strong minimum principle gives $|f_\alpha| > 0$ there. Continuity and connectedness of P_+ show that f_α has one strict sign. Its normalization selects the positive sign, because the weak Neumann equation tested with $\alpha \cdot x$ gives

$$\mu_1(P) \int_P (\alpha \cdot x) f_\alpha(x) \, \mathrm{d}x = \int_P D_\alpha f_\alpha \, \mathrm{d}x = \alpha \cdot Jf_\alpha = 1.$$

If the sign on P_+ were negative, oddness would make the integral on the left negative. Applying (3.6) to the three simple roots proves the last assertion. □

Corollary 3.2 (Central oddness)**.** *For every $a \in \mathbb{R}^3$, one has $f_a(-x) = -f_a(x)$ on $\overline{P}$.*

Proof. The explicit normal orbit in (3.2) gives $P = -P$. Applying (3.4) to $-I$ and using linearity in the parameter gives $f_a(-x) = f_{-a}(x) = -f_a(x)$ on $\overline{P}$. □

4. The exact dodecahedral matrix cone

4.1. **An exact invariant matrix cone for the dodecahedron.** Use the simple roots and dual chamber rays from (1.1)–(1.2), as column vectors, and the corresponding reflections s_i. In particular, $\alpha_i^T b_j = \delta_{ij}$ and $|\alpha_i| = 1$. For a real 3×3 matrix M set

$$\mathcal{P}_i M = \tfrac{1}{2}(M + s_i M s_i), \qquad \mathcal{Q} M = M\left(I - \frac{b_3 b_3^T}{\varphi + 2}\right).$$

Equip matrix space with the Frobenius inner product $\langle L, M\rangle = \operatorname{tr}(L^T M)$. Since $s_i^T = s_i = s_i^{-1}$, conjugation $M \mapsto s_i M s_i$ is a self-adjoint involution for this inner product. Its average with the identity, $\mathcal{P}_i$, is therefore the orthogonal projection onto $\{M : s_i M s_i = M\}$. Likewise, $|b_3|^2 = \varphi + 2$, so $\Pi = I - b_3 b_3^T/(\varphi+2)$ is symmetric and idempotent. Right multiplication by Π is thus self-adjoint and idempotent on matrix space, making $\mathcal{Q}$ the orthogonal projection onto $\{M : M b_3 = 0\}$. Let $e_x = (1,0,0)^T$, $e_y = (0,1,0)^T$, and define the seed matrices

$$A_{ij} = -\alpha_i b_j^T \quad (i \neq j), \qquad V = b_1 b_3^T, \qquad U = b_1 e_x^T, \qquad Y = b_1 e_y^T.$$

Table 1 specifies twenty-eight matrices; operator products in this table act from right to left.

Table 1: Exact generators of the dual matrix cone.

Matrix	Definition
L_0	A_{13}
L_1	A_{23}
L_2	A_{31}
L_3	A_{32}
L_4	V
L_5	U
L_6	$\mathcal{Q}A_{12}$
L_7	$\mathcal{Q}A_{21}$
L_8	$\mathcal{P}_3 A_{23}$
L_9	$\mathcal{Q}Y$
L_{10}	$\mathcal{Q}\mathcal{P}_2 A_{12}$
L_{11}	$\mathcal{Q}\mathcal{P}_1 A_{21}$
L_{12}	$\mathcal{P}_3\mathcal{Q}A_{21}$
L_{13}	$\mathcal{P}_2\mathcal{P}_3 A_{23}$
L_{14}	$\mathcal{Q}\mathcal{P}_3 A_{23}$
L_{15}	$\mathcal{P}_3\mathcal{Q}\mathcal{P}_2 A_{12}$
L_{16}	$\mathcal{P}_3\mathcal{Q}\mathcal{P}_1 A_{21}$
L_{17}	$\mathcal{P}_2\mathcal{P}_3\mathcal{Q}A_{21}$
L_{18}	$\mathcal{P}_1\mathcal{P}_2\mathcal{P}_3 A_{23}$
L_{19}	$\mathcal{Q}\mathcal{P}_3\mathcal{Q}\mathcal{P}_2 A_{12}$
L_{20}	$\mathcal{P}_2\mathcal{P}_3\mathcal{Q}\mathcal{P}_1 A_{21}$
L_{21}	$\mathcal{P}_1\mathcal{P}_2\mathcal{P}_3\mathcal{Q}A_{21}$
L_{22}	$\mathcal{P}_3\mathcal{Q}\mathcal{P}_3\mathcal{Q}\mathcal{P}_2 A_{12}$
L_{23}	$\mathcal{P}_1\mathcal{P}_2\mathcal{P}_3\mathcal{Q}\mathcal{P}_1 A_{21}$
L_{24}	$\mathcal{P}_2\mathcal{P}_1\mathcal{P}_2\mathcal{P}_3\mathcal{Q}A_{21}$
L_{25}	$\mathcal{P}_2\mathcal{P}_1\mathcal{P}_2\mathcal{P}_3\mathcal{Q}\mathcal{P}_1 A_{21}$
L_{26}	$\mathcal{P}_1\mathcal{P}_2\mathcal{P}_1\mathcal{P}_2\mathcal{P}_3\mathcal{Q}\mathcal{P}_1 A_{21}$
L_{27}	$\mathcal{P}_3\mathcal{P}_2\mathcal{P}_1\mathcal{P}_2\mathcal{P}_3\mathcal{Q}\mathcal{P}_1 A_{21}$

Define the closed convex polyhedral cone

$$\mathcal{K}_D = \{M \in \mathbb{R}^{3\times 3} : \langle L_j, M\rangle \geq 0 \text{ for } 0 \leq j \leq 27\}.$$

The terminology "dual generators" refers to the identity

$$\mathcal{K}_D^* := \{L : \langle L, M\rangle \geq 0 \text{ for all } M \in \mathcal{K}_D\} = \operatorname{cone}\{L_0, \ldots, L_{27}\}.$$

Indeed, the conic hull of these finitely many matrices is closed, and its dual is $\mathcal{K}_D$ by definition; the finite-dimensional bipolar theorem gives the displayed equality.

Lemma 4.1 (Invariant dodecahedral matrix cone)**.** *The identity matrix belongs to $\mathcal{K}_D$, and*

$$\mathcal{P}_i(\mathcal{K}_D) \subset \mathcal{K}_D \quad (i = 1, 2, 3), \qquad \mathcal{Q}(\mathcal{K}_D) \subset \mathcal{K}_D.$$

Every $M \in \mathcal{K}_D$ satisfies

$$b_j^T M b_3 \geq 0 \quad (j = 1, 2, 3), \qquad e_y^T M e_y \geq 0, \qquad b_1^T M e_y \geq 0, \qquad b_1^T M b_1 \geq 0,$$
$$b_2^T M(e_y - \varphi e_z) \geq 0, \qquad e_z = (0, 0, 1)^T.$$

Proof. All entries of the generators L_j lie in $\mathbb{Q}(\varphi)$, so the following computations are exact finite algebraic identities. Direct substitution in Table 1, using $\varphi^2 = \varphi + 1$, gives

j	4	5	9	14	19	22
$\operatorname{tr} L_j$	$1 + 2\varphi$	1	$(2 + \varphi)/5$	$(2\varphi - 1)/10$	$1/10$	$1/10$

and $\operatorname{tr} L_j = 0$ for the other twenty-two indices. Since $\langle L_j, I\rangle = \operatorname{tr} L_j$, these nonnegative traces prove $I \in \mathcal{K}_D$.

The complete projector decompositions are recorded in Table 2. Each entry is an identity of the form

$$\mathcal{R} L_j = \sum_{k=0}^{27} c_{jk}^{\mathcal{R}} L_k, \qquad c_{jk}^{\mathcal{R}} \in \mathbb{Q}(\varphi), \quad c_{jk}^{\mathcal{R}} \geq 0, \quad \mathcal{R} \in \{\mathcal{P}_1, \mathcal{P}_2, \mathcal{P}_3, \mathcal{Q}\}.$$

The identities are checked entry by entry from Table 1, reducing products by $\varphi^2 = \varphi + 1$. Their coefficient signs can also be checked using only the rational bounds

$$\frac{8}{5} < \varphi < \frac{5}{3}.$$

These follow by evaluating $t^2 - t - 1$, which is strictly increasing for $t > 1/2$, at the two endpoints: the values are $-1/25$ and $1/9$, respectively. Every coefficient actually appearing in a nonzero table expansion is affine in φ. Substituting $8/5$ for a positive slope and $5/3$ for a negative slope gives a strictly positive rational lower bound in every case; constant coefficients are positive as well. Coefficients absent from an expansion are zero. Consequently, for $M \in \mathcal{K}_D$, self-adjointness gives

$$\langle L_j, \mathcal{R} M\rangle = \langle \mathcal{R} L_j, M\rangle = \sum_k c_{jk}^{\mathcal{R}} \langle L_k, M\rangle \geq 0.$$

This proves invariance under all four projections.

For the seven functionals in the statement, note first that $\langle ab^T, M\rangle = a^T M b$. Their representing matrices have the following explicit decompositions into the same dual generators:

$$\begin{aligned}
b_1 b_3^T &= L_4,\\
b_2 b_3^T &= 2(\varphi-1)(L_0+L_4),\\
b_3 b_3^T &= 4(\varphi-1)L_0 + 2L_1 + (3\varphi-4)L_4,\\
e_y e_y^T &= \frac{3\varphi-4}{5}(L_0+L_4) + \left(\varphi-\tfrac{3}{2}\right)L_6 + (2-\varphi)L_9,\\
b_1 e_y^T &= \frac{2\varphi-1}{5}L_4 + L_9,\\
b_1 b_1^T &= \frac{1+3\varphi}{5}L_4 + L_5 + (1+\varphi)L_9,\\
b_2(e_y-\varphi e_z)^T &= (3-\varphi)L_6 + (4\varphi-2)L_9.
\end{aligned}$$

The same bounds on φ show that all displayed coefficients are positive. Pairing these identities with M proves the seven asserted inequalities. □

Exact verification of the finite certificate. The independent verifier `check_certificate.py` accompanies this article and uses only Python's standard library. It reads the two displayed tables directly from the TEX source, regenerates each L_j from its seed and projector word, checks its trace, and verifies all 112 projector identities and all seven target identities, including the nonnegativity of every coefficient. It also checks the additional tangential and midpoint identities used below. It uses pairs of rational numbers to represent $a+b\varphi$ and the exact multiplication rule

$$(a+b\varphi)(c+d\varphi) = (ac+bd) + (ad+bc+bd)\varphi.$$

For coefficient signs it writes $a+b\varphi = A + B\sqrt{5}$, where $A = a + b/2$ and $B = b/2$. If either A or B is zero, the sign is immediate; if they have the same nonzero sign, that is the sign of their sum. If $AB < 0$, then

$$\operatorname{sgn}(A+B\sqrt{5}) = \operatorname{sgn}(A)\operatorname{sgn}(A^2-5B^2).$$

For nonzero rational A, B, equality $A^2 = 5B^2$ is impossible. Thus the verification uses neither decimal approximations nor a numerical cone-membership test. The displayed tables, independently of that implementation, specify all matrices and all identities needed in the proof.

Table 2: Complete nonnegative decompositions of all projector images. Every coefficient is an exact element of $\mathbb{Q}(\varphi)$.

Image	Nonnegative expansion in the generators
$\mathcal{P}_1 L_0$	0
$\mathcal{P}_2 L_0$	$L_0 + \frac{\varphi}{2}\,L_1$
$\mathcal{P}_3 L_0$	$\frac{\varphi+2}{5}\,L_0 + \frac{1}{2}\,L_6$
$\mathcal{Q} L_0$	0
$\mathcal{P}_1 L_1$	$\frac{\varphi}{2}\,L_0 + L_1$
$\mathcal{P}_2 L_1$	0
$\mathcal{P}_3 L_1$	L_8
$\mathcal{Q} L_1$	0
$\mathcal{P}_1 L_2$	$\frac{\varphi}{2}\,L_3$
$\mathcal{P}_2 L_2$	$\frac{6\varphi+2}{5}\,L_{13} + 2\,L_{17}$
$\mathcal{P}_3 L_2$	0
$\mathcal{Q} L_2$	$\frac{3\varphi+6}{5}\,L_0 + \frac{6\varphi+2}{5}\,L_1 + L_2 + \frac{-\varphi+3}{5}\,L_4$
$\mathcal{P}_1 L_3$	L_3

Image	Nonnegative expansion in the generators
$\mathcal{P}_2 L_3$	$\frac{8\varphi-4}{5}\, L_{13} + 2\varphi\, L_{17}$
$\mathcal{P}_3 L_3$	0
$\mathcal{Q} L_3$	$\big(3\varphi-3\big)\, L_6 + \big(-2\varphi+4\big)\, L_9 + 4\, L_{14}$
$\mathcal{P}_1 L_4$	$L_0 + L_4$
$\mathcal{P}_2 L_4$	L_4
$\mathcal{P}_3 L_4$	$\frac{\varphi+2}{5}\, L_4 + \varphi\, L_9$
$\mathcal{Q} L_4$	0
$\mathcal{P}_1 L_5$	$\frac{5\varphi-6}{2}\, L_6 + \big(-2\varphi+4\big)\, L_9 + \big(2\varphi-2\big)\, L_{11} + 2\, L_{14}$
$\mathcal{P}_2 L_5$	$\frac{-\varphi+3}{4}\, L_5 + \frac{\varphi+2}{4}\, L_9$
$\mathcal{P}_3 L_5$	L_5
$\mathcal{Q} L_5$	L_5
$\mathcal{P}_1 L_6$	0
$\mathcal{P}_2 L_6$	L_{10}
$\mathcal{P}_3 L_6$	$\frac{2}{5}\, L_0 + \frac{-\varphi+3}{5}\, L_6$
$\mathcal{Q} L_6$	L_6
$\mathcal{P}_1 L_7$	L_{11}
$\mathcal{P}_2 L_7$	0
$\mathcal{P}_3 L_7$	L_{12}
$\mathcal{Q} L_7$	L_7
$\mathcal{P}_1 L_8$	$\frac{3\varphi+1}{10}\, L_0 + \frac{\varphi}{4}\, L_6 + L_8$
$\mathcal{P}_2 L_8$	L_{13}
$\mathcal{P}_3 L_8$	L_8
$\mathcal{Q} L_8$	L_{14}
$\mathcal{P}_1 L_9$	$\frac{\varphi-1}{2}\, L_6 + L_9$
$\mathcal{P}_2 L_9$	$\frac{1}{4}\, L_5 + \frac{\varphi+1}{4}\, L_9$
$\mathcal{P}_3 L_9$	$\frac{\varphi-1}{5}\, L_4 + \frac{-\varphi+3}{5}\, L_9$
$\mathcal{Q} L_9$	L_9
$\mathcal{P}_1 L_{10}$	$\frac{\varphi-1}{2}\, L_{11}$
$\mathcal{P}_2 L_{10}$	L_{10}
$\mathcal{P}_3 L_{10}$	L_{15}
$\mathcal{Q} L_{10}$	L_{10}
$\mathcal{P}_1 L_{11}$	L_{11}
$\mathcal{P}_2 L_{11}$	$\frac{\varphi-1}{2}\, L_{10}$
$\mathcal{P}_3 L_{11}$	L_{16}
$\mathcal{Q} L_{11}$	L_{11}
$\mathcal{P}_1 L_{12}$	L_{16}
$\mathcal{P}_2 L_{12}$	L_{17}
$\mathcal{P}_3 L_{12}$	L_{12}
$\mathcal{Q} L_{12}$	$\frac{3}{5}\, L_0 + \frac{\varphi}{5}\, L_1 + \frac{-\varphi+2}{5}\, L_4 + L_{12}$
$\mathcal{P}_1 L_{13}$	L_{18}
$\mathcal{P}_2 L_{13}$	L_{13}
$\mathcal{P}_3 L_{13}$	$\frac{1}{4}\, L_8$
$\mathcal{Q} L_{13}$	$\frac{-3\varphi+9}{10}\, L_0 + \frac{6\varphi-3}{20}\, L_1 + \frac{-4\varphi+7}{10}\, L_4 + L_{13}$
$\mathcal{P}_1 L_{14}$	$\frac{\varphi}{4}\, L_6 + L_{14}$
$\mathcal{P}_2 L_{14}$	$\frac{-3\varphi+9}{10}\, L_0 + \frac{6\varphi-3}{20}\, L_1 + \frac{-4\varphi+7}{10}\, L_4 + L_{13}$

Image	Nonnegative expansion in the generators
$\mathcal{P}_3L_{14}$	$\frac{1}{5}\,L_1 + \frac{-3\varphi+9}{20}\,L_6 + \frac{3\varphi-4}{10}\,L_9 + \frac{\varphi+2}{5}\,L_{14}$
$\mathcal{Q}L_{14}$	L_{14}
$\mathcal{P}_1L_{15}$	$\frac{\varphi-1}{2}\,L_{16}$
$\mathcal{P}_2L_{15}$	$\frac{\varphi+1}{10}\,L_0 + \frac{2\varphi+1}{20}\,L_1 + \frac{-4\varphi+17}{20}\,L_{10} + \frac{2\varphi-1}{5}\,L_{13}$
$\mathcal{P}_3L_{15}$	L_{15}
$\mathcal{Q}L_{15}$	L_{19}
$\mathcal{P}_1L_{16}$	L_{16}
$\mathcal{P}_2L_{16}$	L_{20}
$\mathcal{P}_3L_{16}$	L_{16}
$\mathcal{Q}L_{16}$	$\frac{-\varphi+5}{10}\,L_0 + \frac{\varphi}{5}\,L_1 + \frac{-\varphi+2}{5}\,L_4 + L_{16}$
$\mathcal{P}_1L_{17}$	L_{21}
$\mathcal{P}_2L_{17}$	L_{17}
$\mathcal{P}_3L_{17}$	$\frac{1}{4}\,L_{12}$
$\mathcal{Q}L_{17}$	$\frac{2\varphi-3}{4}\,L_5 + \frac{1}{5}\,L_6 + \frac{\varphi-1}{4}\,L_9 + \frac{17\varphi-26}{10}\,L_{10} + \frac{4\varphi-4}{5}\,L_{14} + \frac{-16\varphi+28}{5}\,L_{19}$
$\mathcal{P}_1L_{18}$	L_{18}
$\mathcal{P}_2L_{18}$	$\frac{\varphi+2}{40}\,L_0 + \frac{3\varphi+1}{80}\,L_1 + \frac{\varphi}{8}\,L_{10} + \frac{-\varphi+8}{10}\,L_{13} + \frac{1}{4}\,L_{17}$
$\mathcal{P}_3L_{18}$	$\frac{3\varphi+1}{40}\,L_0 + \frac{\varphi}{16}\,L_6 + \frac{1}{4}\,L_8$
$\mathcal{Q}L_{18}$	$\frac{-\varphi+3}{4}\,L_6 + \frac{2\varphi-3}{4}\,L_9 + \frac{\varphi}{2}\,L_{14}$
$\mathcal{P}_1L_{19}$	$\frac{5\varphi-6}{20}\,L_0 + \frac{1}{10}\,L_1 + \frac{2\varphi-3}{10}\,L_4 + \frac{\varphi-1}{2}\,L_{16}$
$\mathcal{P}_2L_{19}$	$\frac{\varphi+1}{20}\,L_6 + \frac{1}{4}\,L_{10} + \frac{\varphi}{5}\,L_{14} + \frac{-\varphi+3}{5}\,L_{19}$
$\mathcal{P}_3L_{19}$	L_{22}
$\mathcal{Q}L_{19}$	L_{19}
$\mathcal{P}_1L_{20}$	L_{23}
$\mathcal{P}_2L_{20}$	L_{20}
$\mathcal{P}_3L_{20}$	$\frac{22\varphi-16}{145}\,L_{15} + \frac{3\varphi+11}{145}\,L_{16} + \frac{-4\varphi+24}{29}\,L_{27}$
$\mathcal{Q}L_{20}$	$\frac{-\varphi+5}{10}\,L_0 + \frac{4\varphi-1}{20}\,L_1 + \frac{-\varphi+2}{5}\,L_4 + L_{20}$
$\mathcal{P}_1L_{21}$	L_{21}
$\mathcal{P}_2L_{21}$	L_{24}
$\mathcal{P}_3L_{21}$	$\frac{1}{4}\,L_{16}$
$\mathcal{Q}L_{21}$	$\frac{-\varphi+3}{4}\,L_6 + \frac{11\varphi-13}{20}\,L_9 + \frac{1}{4}\,L_{11} + \frac{6\varphi-3}{10}\,L_{14}$
$\mathcal{P}_1L_{22}$	$\frac{-\varphi+2}{80}\,L_4 + \frac{1}{16}\,L_5 + \frac{3\varphi-4}{80}\,L_9 + \frac{2\varphi-1}{16}\,L_{12} + \frac{\varphi}{8}\,L_{15} + \frac{-\varphi+2}{4}\,L_{22}$
$\mathcal{P}_2L_{22}$	$\frac{1}{25}\,L_0 + \frac{1}{200}\,L_1 + \frac{11\varphi+11}{200}\,L_6 + \frac{2\varphi-1}{50}\,L_8 + \frac{2\varphi-1}{20}\,L_{10} + \frac{9\varphi+1}{50}\,L_{14} + \frac{-11\varphi+33}{50}\,L_{19} + \frac{\varphi-1}{5}\,L_{20}$
$\mathcal{P}_3L_{22}$	L_{22}
$\mathcal{Q}L_{22}$	$\frac{13\varphi-19}{100}\,L_0 + \frac{\varphi+2}{50}\,L_1 + \frac{2\varphi-3}{20}\,L_4 + L_{22}$
$\mathcal{P}_1L_{23}$	L_{23}
$\mathcal{P}_2L_{23}$	L_{25}
$\mathcal{P}_3L_{23}$	$\frac{3\varphi+1}{100}\,L_0 + \frac{\varphi}{40}\,L_6 + \frac{1}{10}\,L_8 + \frac{-\varphi+3}{5}\,L_{16}$
$\mathcal{Q}L_{23}$	$\frac{-\varphi+4}{8}\,L_0 + \frac{4\varphi-1}{20}\,L_1 + \frac{-\varphi+2}{5}\,L_4 + L_{23}$
$\mathcal{P}_1L_{24}$	$\frac{\varphi}{32}\,L_3 + \frac{-\varphi+2}{16}\,L_{16} + \frac{-\varphi+3}{10}\,L_{18} + \frac{\varphi+1}{4}\,L_{23}$
$\mathcal{P}_2L_{24}$	L_{24}
$\mathcal{P}_3L_{24}$	$\frac{1}{10}\,L_8 + \frac{-\varphi+3}{80}\,L_{12} + \frac{3\varphi-4}{40}\,L_{15} + \frac{\varphi+2}{20}\,L_{16}$
$\mathcal{Q}L_{24}$	$\frac{11\varphi-13}{80}\,L_5 + \frac{3\varphi+3}{40}\,L_6 + \frac{9\varphi-2}{80}\,L_9 + \frac{7\varphi-11}{40}\,L_{10} + \frac{3\varphi}{10}\,L_{14} + \frac{-3\varphi+9}{10}\,L_{19}$
$\mathcal{P}_1L_{25}$	L_{26}
$\mathcal{P}_2L_{25}$	L_{25}

Image	Nonnegative expansion in the generators
$\mathcal{P}_3 L_{25}$	L_{27}
$\mathcal{Q} L_{25}$	$\frac{-\varphi+4}{8} L_0 + \frac{3\varphi-1}{16} L_1 + \frac{-\varphi+2}{5} L_4 + L_{25}$
$\mathcal{P}_1 L_{26}$	L_{26}
$\mathcal{P}_2 L_{26}$	$\frac{\varphi+2}{80} L_0 + \frac{3\varphi+1}{160} L_1 + \frac{\varphi-1}{8} L_{13} + \frac{13\varphi-20}{16} L_{20} + \frac{-2\varphi+5}{2} L_{25}$
$\mathcal{P}_3 L_{26}$	$\frac{19\varphi+8}{400} L_0 + \frac{6\varphi+1}{160} L_6 + \frac{\varphi+5}{40} L_8 + \frac{-3\varphi+9}{20} L_{16}$
$\mathcal{Q} L_{26}$	$\frac{-22\varphi+79}{160} L_0 + \frac{3\varphi-1}{16} L_1 + \frac{-\varphi+2}{5} L_4 + L_{26}$
$\mathcal{P}_1 L_{27}$	$\frac{19\varphi+8}{400} L_0 + \frac{6\varphi+1}{160} L_6 + \frac{\varphi+5}{40} L_8 + \frac{-3\varphi+9}{20} L_{16}$
$\mathcal{P}_2 L_{27}$	$\frac{9\varphi+3}{400} L_0 + \frac{3\varphi}{160} L_6 + \frac{3}{40} L_8 + \frac{9\varphi-12}{40} L_{15} + \frac{1}{4} L_{25}$
$\mathcal{P}_3 L_{27}$	L_{27}
$\mathcal{Q} L_{27}$	$\frac{-29\varphi+56}{160} L_0 + \frac{3\varphi-4}{40} L_1 + \frac{-13\varphi+23}{80} L_4 + L_{27}$

5. Matrix heat flow and invariant convex cones

Use the Frobenius inner product and norm from Section 4. For a matrix-valued function, $|DM|^2$ denotes the sum of the squares of the first derivatives of all its entries. We first justify the weak boundary problem used below; this avoids any assumption of smoothness of eigenfunctions at polyhedral edges.

5.1. **Boundary subspaces and approximation.** Let D be the interior of a bounded full-dimensional convex polytope in $\mathbb{R}^3$. To each open facet Γ assign an orthogonal projection R_Γ of a finite dimensional Euclidean space V onto a linear subspace S_Γ. Set

$$\mathcal{V}_R(D) = \{X \in H^1(D;V) : X|_\Gamma \in S_\Gamma \text{ for every facet } \Gamma\}.$$

The trace conditions are understood almost everywhere. Continuity of each facet trace map and closedness of S_Γ make this a closed subspace of $H^1(D;V)$. It contains $C_c^\infty(D;V)$ and is therefore dense in $L^2(D;V)$. On this Hilbert space consider the nonnegative symmetric form

$$\mathfrak{b}_R[X,Y] = \int_D \langle DX, DY\rangle, \qquad \operatorname{dom}\mathfrak{b}_R = \mathcal{V}_R(D). \tag{5.1}$$

Its form norm, $\left(\|X\|_2^2 + \mathfrak{b}_R[X,X]\right)^{1/2}$, is exactly the H^1 norm. Thus the form is closed, and its associated nonnegative self-adjoint operator is denoted by B_R.

Lemma 5.1 (Cone preservation)**.** *Let $K \subset V$ be a closed convex cone such that*

$$R_\Gamma K \subset K \qquad \text{for every facet } \Gamma. \tag{5.2}$$

If $X_0 \in L^2(D;V)$ and $X_0(x) \in K$ almost everywhere, then $e^{-tB_R}X_0(x) \in K$ almost everywhere for every $t \ge 0$.

Proof. Let π_K be the metric projection onto K. We first check that π_K preserves each boundary subspace. If $x \in S_\Gamma$ and $z = \pi_K x$, then $R_\Gamma z \in K$ by (5.2), and orthogonality gives

$$|x - z|^2 = |x - R_\Gamma z|^2 + |(I - R_\Gamma)z|^2.$$

The minimizing property of z implies $(I - R_\Gamma)z = 0$. Thus $\pi_K(S_\Gamma) \subset S_\Gamma$.

The map π_K is 1-Lipschitz and fixes zero. The Sobolev chain and trace rules give

$$|\partial_\ell(\pi_K X)| \le |\partial_\ell X| \quad \text{almost everywhere}, \qquad \operatorname{Tr}_\Gamma(\pi_K X) = \pi_K(\operatorname{Tr}_\Gamma X).$$

Together with preservation of the face subspaces, these imply

$$\pi_K X \in \mathcal{V}_R(D), \qquad \mathfrak{b}_R[\pi_K X, \pi_K X] \le \mathfrak{b}_R[X,X] \quad (X \in \mathcal{V}_R(D)).$$

For $x \in V$, $z = \pi_K x$, and $y \in K$, the minimizing property of z, applied to $z + s(y - z)$ as $s \downarrow 0$, gives $\langle x - z, y - z\rangle \le 0$. Expanding $|x - y|^2$ therefore yields

$$|z - y|^2 + |x - z|^2 \le |x - y|^2.$$

For a K-valued $Y \in L^2(D; V)$, integration gives in particular $\|\pi_K X - Y\|_2^2 \le \|X - Y\|_2^2$. For $h > 0$, the resolvent $(I + hB_R)^{-1}Y$ is the unique minimizer of

$$X \longmapsto \|X - Y\|_{L^2}^2 + h\mathfrak{b}_R[X, X] \quad \text{on } \mathcal{V}_R(D).$$

Projection onto K cannot increase this functional, so uniqueness implies that its minimizer takes values in K almost everywhere. All resolvents thus preserve the cone of K-valued functions. This cone is closed in L^2: from an L^2-convergent sequence one can extract a subsequence converging almost everywhere, and then use closedness of K. For $t > 0$, the spectral theorem gives

$$e^{-tB_R}Y = \lim_{m\to\infty}(I + tB_R/m)^{-m}Y \quad \text{in } L^2.$$

Indeed, the scalar multipliers $(1 + t\lambda/m)^{-m}$ converge to $e^{-t\lambda}$ and are bounded by one for $\lambda \ge 0$, so dominated convergence applies to the spectral measure. Closedness of the cone now proves the assertion for $t > 0$; the case $t = 0$ is the hypothesis. □

5.2. **The derivative of the coordinate heat flow.** Return to the regular dodecahedron P, its reflection group W, and the flag chamber T. Put $n = b_3$ and use the matrix projections $\mathcal{P}_i$ and $\mathcal{Q}$ from Section 4. The three mirror facets of T lie in $\alpha_i \cdot x = 0$. Its fourth facet lies in the physical face Γ of P and has normal parallel to n. Assign the projection $\mathcal{P}_i$ to the ith mirror facet and $\mathcal{Q}$ to the physical facet. Let B denote the operator of (5.1) for these assignments on T.

Write $A = -\Delta_{\mathrm{N},P}$. Apply the scalar Neumann heat semigroup to the three coordinate functions and define

$$U(t, x) = \big(e^{-tA}x_1, e^{-tA}x_2, e^{-tA}x_3\big)^T, \qquad M(t, x) = D_x U(t, x). \tag{5.3}$$

Here and below the convention is $M_{ij} = \partial_j U_i$.

Proposition 5.2 (Matrix boundary problem)**.** *For $t > 0$, the restriction of $M(t)$ to T is*

$$M(t)|_T = e^{-tB} I. \tag{5.4}$$

Here I is the constant matrix field in $L^2(T; \mathbb{R}^{3\times 3})$, and the equality is in this space.

Proof. The invariance of the scalar form under W and the initial condition $U(0, x) = x$ imply

$$U(t, wx) = wU(t, x), \qquad M(t, wx) = wM(t, x)w^T \quad (w \in W). \tag{5.5}$$

For positive time the spectral theorem gives $U(t)$ in every power of the operator domain and, on each interval $0 < \delta \le t \le \tau$, $U \in C^1([\delta, \tau]; \operatorname{dom} A)$ in the graph norm. The inclusion $\operatorname{dom} A \subset H^2(P)$ established in Section 2 is continuous by the closed graph theorem: convergence in either norm implies convergence to the same L^2 limit. Consequently,

$$M \in C([\delta, \tau]; H^1(P)) \cap C^1([\delta, \tau]; L^2(P)), \qquad \partial_t M = D\partial_t U = D\Delta U.$$

In particular, $\Delta U = -AU \in H^2(P)$ at positive times.

We use the polarized flat Gaffney identity on P. If u is a positive-time scalar Neumann heat solution and $X \in H^1(P; \mathbb{R}^3)$ is tangent to every facet, polarizing the tangent-field identity in the proof of Lemma 2.2 with ∇u and X, and using the vanishing antisymmetric derivatives of ∇u, gives

$$\int_P D\nabla u : DX = \int_P \Delta u \operatorname{div} X = -\int_P \nabla\Delta u \cdot X. \tag{5.6}$$

The last equality is the weak Green formula: $\Delta u \in H^1(P)$ for positive time, and the normal trace of X vanishes. Since $\partial_t \nabla u = \nabla\Delta u$, applying (5.6) to the three components of U yields

$$\int_P \langle \partial_t M, H\rangle + \int_P \langle DM, DH\rangle = 0 \tag{5.7}$$

for every $H \in H^1(P;\mathbb{R}^{3\times 3})$ such that $H\nu = 0$ on each physical facet. The H^1 validity of the polarized identity is already supplied by Lemma 2.2.

The Neumann trace of U gives $Mn = 0$ on the physical facet of T. On its ith mirror facet, (5.5) gives $M = s_i M s_i$. Thus $M(t)|_T \in \mathcal{V}_R(T)$. For an arbitrary $H \in \mathcal{V}_R(T)$, define an extension to all chambers by

$$\widetilde{H}(wx) = wH(x)w^T \qquad (x \in T,\ w \in W).$$

The traces from adjacent chambers agree, because on a shared mirror $H = s_i H s_i$. More explicitly, on the interface between wT and ws_iT the traces are wHw^T and $ws_iHs_iw^T$, respectively, at the same base point of the ith mirror. These coincide. In the distributional derivatives over this finite polyhedral partition the interface terms therefore cancel, giving $\widetilde{H} \in H^1(P)$. On a physical facet its normal is the corresponding orthogonal image of n, and $\widetilde{H}(wx)(wn) = wH(x)n = 0$. Hence $\widetilde{H}$ is an admissible test in (5.7).

All chamber integrals in that equation are equal, by equivariance of M and $\widetilde{H}$ and orthogonality of the action on values and spatial variables. Dividing by $|W|$ yields

$$(\partial_t M, H)_{L^2(T)} + \mathfrak{b}_R[M, H] = 0 \qquad (H \in \mathcal{V}_R(T)). \tag{5.8}$$

For each positive time, $M(t)|_T$ belongs to the form domain and (5.8) represents its form functional by $-\partial_t M(t) \in L^2(T)$. The definition of the associated operator therefore gives

$$M(t)|_T \in \operatorname{dom} B, \qquad BM(t) = -\partial_t M(t).$$

Fix $0 < \delta < t$. Uniqueness for this strong evolution gives $M(t) = e^{-(t-\delta)B}M(\delta)$ on T. The scalar form domain of A is $H^1(P)$, and each coordinate function belongs to that domain. Strong continuity of its heat semigroup in the form norm gives $D(e^{-\delta A}x_j) \to e_j$ in $L^2(P)$ as $\delta \downarrow 0$. Thus $M(\delta) \to I$ in $L^2(T)$. For fixed $t > 0$, contractivity gives

$$\|e^{-(t-\delta)B}M(\delta) - e^{-tB}I\|_2 \le \|M(\delta) - I\|_2 + \|(e^{-(t-\delta)B} - e^{-tB})I\|_2 \longrightarrow 0.$$

Strong continuity of e^{-tB} thus proves (5.4). □

5.3. **Passage to the first eigenspace.** Write $\mu = \mu_1(P) > 0$ and retain the canonical eigenmap F from Section 3.

Proposition 5.3 (A cone criterion for the eigenmap)**.** *Let $K \subset \mathbb{R}^{3\times 3}$ be a closed convex cone such that*

$$I \in K, \qquad \mathcal{P}_i K \subset K\ (i = 1, 2, 3), \qquad \mathcal{Q}K \subset K.$$

Then $DF(x) \in K$ for every $x \in T$.

Proof. By Proposition 5.2 and Lemma 5.1, $M(t, x) \in K$ almost everywhere on T for every $t > 0$. We check the sign of its leading spectral coefficient explicitly. The bilinear form $(a, b) \mapsto \int_P f_a f_b$ is positive definite because $a \mapsto f_a$ is an isomorphism. Covariance and invariance of volume make it W-invariant. Its representing symmetric matrix therefore commutes with W, so every eigenspace of that matrix is W-invariant. Irreducibility forces the matrix to be κI with $\kappa > 0$. Thus

$$\int_P f_a f_b = \kappa\, a \cdot b \qquad (\kappa > 0).$$

Testing the weak eigenvalue equation for f_{e_i} with x_j gives

$$\mu \int_P x_j f_{e_i} = \int_P \partial_j f_{e_i} = \delta_{ij}.$$

The orthogonal projection of x_j onto $E_1(P)$ is consequently $(\mu\kappa)^{-1} f_{e_j}$. The coordinate functions have mean zero because $P = -P$. Let $\lambda_* > \mu$ be the next distinct Neumann eigenvalue and put

$r_j = x_j - (\mu\kappa)^{-1} f_{e_j}$. Then $r_j \in H^1(P)$ and all its spectral components have eigenvalues at least λ_*. Since the Neumann form norm is the H^1 norm, the spectral theorem gives

$$\|e^{\mu t} e^{-tA} r_j\|_{H^1(P)} \le e^{-(\lambda_* - \mu)t} \|r_j\|_{H^1(P)} \longrightarrow 0.$$

Consequently,

$$e^{\mu t} U(t) \longrightarrow (\mu\kappa)^{-1} F \quad \text{in } H^1(P), \qquad e^{\mu t} M(t) \longrightarrow (\mu\kappa)^{-1} DF \quad \text{in } L^2(P).$$

The L^2-closedness of the cone of K-valued functions was proved in Lemma 5.1. Conicity and the positive coefficient $(\mu\kappa)^{-1}$ therefore imply $DF \in K$ almost everywhere on T. Finally, DF is continuous in the interior of P. If its value at some point of T were outside the closed set K, it would remain outside on a neighborhood of positive measure, contradicting the almost-everywhere assertion. □

6. Strict inequalities on mirror strata and the hot spots theorem

6.1. **Strictness at intersections of mirrors.**

For $K \subset \mathbb{R}^{3\times 3}$, its dual cone is

$$K^* = \{L : \langle L, M\rangle \ge 0 \text{ for every } M \in K\}.$$

This is a closed convex cone, being an intersection of closed half-spaces. If $\mathcal{P}_i K \subset K$, self-adjointness gives $\langle \mathcal{P}_i H, M\rangle = \langle H, \mathcal{P}_i M\rangle \ge 0$ for $H \in K^*$ and $M \in K$, so $\mathcal{P}_i K^* \subset K^*$.

Lemma 6.1 (Strictness up to interior mirror strata)**.** *Suppose $K \subset \mathbb{R}^{3\times 3}$, $DF \in K$ on T, and $\mathcal{P}_i K \subset K$ for $i = 1, 2, 3$. If $L \in K^*$ and $\operatorname{tr} L > 0$, then*

$$\langle L, DF(q)\rangle > 0 \qquad (q \in P \cap \overline{C}). \tag{6.1}$$

Here q is allowed to lie on several symmetry mirrors, but is an interior point of the full polytope P.

Proof. For any $H \in K^*$, differentiating the interior eigenvalue equations for the analytic components of F shows that $g_H = \langle H, DF\rangle$ satisfies $-\Delta g_H = \mu g_H$ in P. It is nonnegative in T by the dual-cone condition. If $\operatorname{tr} H > 0$, it cannot vanish identically on T: analyticity would then make it vanish on the connected set P, contrary to (3.5) and

$$\int_P g_H = \left\langle H, \int_P DF \right\rangle = \operatorname{tr} H > 0.$$

On the connected chamber T, one has $\Delta g_H = -\mu g_H \le 0$. The strong minimum principle for superharmonic functions therefore gives $g_H > 0$ there.

Now fix $q \in P \cap \overline{C}$, and let $J = \{i : \alpha_i \cdot q = 0\}$. If J is empty, the preceding argument applies directly. Otherwise let W_J be the subgroup generated by s_j, $j \in J$, and let P_J be the orthogonal projection of matrix space onto

$$\mathcal{S}_J = \{H : wHw^T = H \text{ for every } w \in W_J\}.$$

Cyclic products of the orthogonal projections $\mathcal{P}_j$, $j \in J$, converge to P_J. Indeed, their common fixed space is $\mathcal{S}_J$, since being fixed under the generating reflections is equivalent to being fixed under W_J. Each $\mathcal{P}_j$ is the identity on $\mathcal{S}_J$ and, by self-adjointness, preserves $\mathcal{S}_J^\perp$. If one full cycle preserved the norm of a nonzero vector in $\mathcal{S}_J^\perp$, every projection in the cycle would preserve its norm. Orthogonality then forces each projection to fix that vector, putting it in $\mathcal{S}_J$, a contradiction. Compactness of the unit sphere in $\mathcal{S}_J^\perp$ therefore bounds the cycle norm there by some $\rho < 1$. Iterating the cycle is the identity on $\mathcal{S}_J$ and tends to zero on its orthogonal complement, which proves convergence to P_J in operator norm. It follows that

$$L' = P_J L \in K^*, \qquad \operatorname{tr} L' = \operatorname{tr} L > 0,$$

because the cycle preserves K^*, this cone is closed, and $\operatorname{tr}(\mathcal{P}_j H) = \operatorname{tr} H$ by cyclicity of the trace.

The function $g_{L'}$ is invariant under W_J, by covariance of DF and the definition of P_J. By the stabilizer and local chamber decomposition established in Section 3, a sufficiently small ball

$B(q,r) \subset P$ is covered by $w(B(q,r)\cap\overline{C})$, $w \in W_J$. Continuity gives $g_{L'} \geq 0$ on $P\cap\overline{C}$, so invariance gives this inequality on the whole ball. There $\Delta g_{L'} = -\mu g_{L'} \leq 0$, and $g_{L'} > 0$ on the nonempty set $B(q,r)\cap T$ by the first part of the proof. The strong minimum principle therefore gives $g_{L'}(q) > 0$.

Every element of W_J fixes q, so equivariance gives $DF(q) \in \mathcal{S}_J$. Orthogonality of P_J now yields

$$\langle L, DF(q)\rangle = \langle P_J L, DF(q)\rangle = g_{L'}(q) > 0.$$

□

6.2. **Application of the exact dodecahedral cone.**

Proposition 6.2 (A common direction of increase)**.** *For every nonzero $a \in \overline{C}$,*

$$D_{b_3} f_a(q) > 0 \qquad (q \in P \cap \overline{C}). \tag{6.2}$$

Proof. Lemma 4.1 and Proposition 5.3 give $DF \in \mathcal{K}_D$ on T. Write $a = \sum_j t_j b_j$, where $t_j = \alpha_j \cdot a \geq 0$ and at least one t_j is positive. The cone lemma gives $b_j b_3^T \in \mathcal{K}_D^*$, so

$$L = a b_3^T = \sum_j t_j b_j b_3^T \in \mathcal{K}_D^*.$$

Its trace is $a \cdot b_3 > 0$, since the three numbers $b_j \cdot b_3$ are $2\varphi + 1$, $2\varphi^2$, and $\varphi + 2$, respectively. Lemma 6.1 now gives

$$D_{b_3} f_a(q) = \langle L, DF(q)\rangle > 0 \qquad (q \in P \cap \overline{C}).$$

This is (6.2) and Theorem 1.2. □

Theorem 6.3 (Hot spots on the regular dodecahedron)**.** *Every nonzero real first positive Neumann eigenfunction on a regular dodecahedron attains its global maximum and global minimum on the boundary only.*

Proof. It remains to prove the extremum assertion in Theorem 1.1. Continuity on $\overline{P}$ guarantees that the extrema exist. Write $u = f_a$ with $a \neq 0$, and choose $w \in W$ such that $wa \in \overline{C}$. Covariance gives $f_a \circ w^{-1} = f_{wa}$, and w preserves P, so it suffices to treat $a \in \overline{C}$. Covariance and linearity in the parameter give, for each simple root,

$$f_a(s_i x) - f_a(x) = -2(a \cdot \alpha_i) f_{\alpha_i}(x). \tag{6.3}$$

By Proposition 3.1 and oddness under s_i, $f_{\alpha_i}(x)$ has the strict sign of $\alpha_i \cdot x$ in P.

Suppose that $x \in P$ is a global maximum of f_a. If $\alpha_i \cdot x < 0$, the right side of (6.3) is nonnegative and is strictly positive when $a \cdot \alpha_i > 0$. Thus maximality forces $a \cdot \alpha_i = 0$ for any such reflection. Reflecting x across this wall preserves both a and the maximal value of f_a.

Repeat this operation whenever a simple-root coordinate is negative. The finite-orbit termination argument in Section 3 applies to this sequence. Every step stays in P and preserves the global maximum of the same function f_a. At termination we obtain $y \in P \cap \overline{C}$ which is still a global maximum of the same function f_a. Proposition 6.2 gives $D_{b_3} f_a(y) > 0$. Since y is interior to P, for sufficiently small $s > 0$ one has $y + s b_3 \in P$ and $f_a(y + s b_3) > f_a(y)$, contradicting maximality. Thus no interior global maximum exists. By central oddness (Corollary 3.2), an interior global minimum would give an interior global maximum at its antipode. This excludes interior global minima as well. □

7. Locations of the global extrema

7.1. **Positivity at physical edges and vertices.**

Lemma 7.1 (Local Neumann positivity on a Lipschitz boundary)**.** *Let $D \subset \mathbb{R}^3$ be a Lipschitz domain, let $q \in \partial D$, and let $u \in H^1(D \cap U)$ satisfy, in a neighborhood U of q,*

$$\int_{D\cap U} \nabla u \cdot \nabla \eta = \lambda \int_{D\cap U} u\eta \qquad (\lambda \geq 0)$$

for all H^1 test functions compactly supported in U relative to $\overline{D}$, whose traces on ∂D need not vanish. Then u has a locally Hölder continuous representative up to ∂D near q. If, in addition, $D \cap U$ is connected, $u \geq 0$ there, and u is not identically zero there, this representative is strictly positive at q. In particular, no smoothness of an edge or vertex is required.

Proof. After a rigid change of coordinates, a sufficiently small boundary patch has the form $x_3 > h(x_1, x_2)$, where h is Lipschitz. Write

$$\Phi(y', t) = (y', h(y') + t), \qquad v(y', t) = u(\Phi(y', t)), \quad t > 0.$$

The map Φ is bi-Lipschitz and $\det D\Phi = 1$ almost everywhere. The transformed equation on a half-cylinder is

$$\int A\nabla v \cdot \nabla\zeta = \lambda \int v\zeta, \qquad A = (D\Phi)^{-1}(D\Phi)^{-T} = \begin{pmatrix} I_2 & -\nabla h \\ -\nabla h^T & 1 + |\nabla h|^2 \end{pmatrix}.$$

Here A is measurable, bounded, symmetric, and uniformly elliptic, with constants depending only on the Lipschitz constant of h. The weak formulation allows arbitrary traces on $t = 0$.

Let $R = \operatorname{diag}(1, 1, -1)$. Extend v evenly and reflect the coefficient matrix by

$$\widetilde{v}(y', t) = v(y', |t|), \qquad \widetilde{A}(y', t) = \begin{cases} A(y', t), & t > 0, \\ RA(y', -t)R, & t < 0. \end{cases}$$

These extensions are defined on a full cylinder. Testing the half-cylinder equation with $\zeta(y', t) + \zeta(y', -t)$ shows directly that

$$-\operatorname{div}(\widetilde{A}\nabla\widetilde{v}) = \lambda\widetilde{v}$$

weakly across $t = 0$. In particular no surface source appears: its absence is precisely the homogeneous Neumann condition encoded in the original weak formulation. The extended coefficient matrix has the same ellipticity bounds as A.

For completeness the zeroth-order term can be removed by adding one variable. On a product cylinder in $\mathbb{R}^4$, set

$$z(y, \tau) = \widetilde{v}(y)\cosh(\sqrt{\lambda}\,\tau), \qquad \mathscr{A}(y, \tau) = \operatorname{diag}(\widetilde{A}(y), 1).$$

The extended equation implies directly, in the weak sense,

$$\operatorname{div}_{y,\tau}(\mathscr{A}\nabla z) = 0.$$

Indeed the y divergence is $-\lambda\widetilde{v}\cosh(\sqrt{\lambda}\,\tau)$, and the last second derivative cancels it. Fubini's theorem and integration by parts in τ justify this calculation for the H^1_{loc} function z. The coefficient matrix is bounded and uniformly elliptic. Interior Hölder regularity for this homogeneous divergence-form equation applies without a sign assumption; see [2, Chapter 8]. The H^1 trace of z on $\tau = 0$ is $\widetilde{v}$, so its continuous restriction there represents $\widetilde{v}$. Transferring back by the bi-Lipschitz map Φ proves the continuity assertion. If $u \geq 0$, then $z \geq 0$ as well. The interior Harnack inequality makes the continuous representative of z either positive or identically zero on each connected product cylinder. Thus $u(q) = 0$ would force u to vanish on an interior neighborhood next to q. In the connected set $D \cap U$, interior regularity and $u \geq 0$ give $\Delta u = -\lambda u \leq 0$. The strong minimum principle then forces $u \equiv 0$ there, contradicting the hypothesis. This proves the positivity assertion. □

Corollary 7.2 (Strict root signs on the closed solid)**.** *For every unit reflection normal α,*

$$f_\alpha(q) > 0 \qquad (q \in \overline{P},\ \alpha \cdot q > 0).$$

It vanishes on $\overline{P} \cap \{\alpha \cdot q = 0\}$ and is negative on $\overline{P} \cap \{\alpha \cdot q < 0\}$.

Proof. Proposition 3.1 gives the interior sign, and continuity gives nonnegativity on the closure of the positive half-solid. Given a physical boundary point q with $\alpha \cdot q > 0$, choose a sufficiently small ball about q disjoint from the reflection plane. On this patch the weak boundary condition is purely Neumann, and the solution is positive on its interior. Lemma 7.1 therefore gives $f_\alpha(q) > 0$. Oddness under the reflection gives the zero and negative assertions. □

7.2. **Boundary regions determined by the parameter.**

Theorem 7.3 (Localization for every coefficient vector)**.** *For $a \in \overline{C} \setminus \{0\}$, let $\mathcal{M}(a) = \operatorname{Argmax}_{\overline{P}} f_a$ and $\Updownarrow(a) = \operatorname{Argmin}_{\overline{P}} f_a$. Then*

$$\varnothing \neq \mathcal{M}(a) \subset \bigcup_{w \in W_a} w\Delta, \qquad \Updownarrow(a) = -\mathcal{M}(a), \qquad W_a = \{w \in W : wa = a\}. \tag{7.1}$$

In particular, a strictly dominant parameter has all maxima in Δ and all minima in $-\Delta$. A positive face-axis parameter $a = \lambda b_3$ has all maxima in Γ and all minima in $-\Gamma$.

Proof. The reflection identity (6.3) holds for any unit reflection normal α, by the same covariance and parameter linearity, and extends to $\overline{P}$ by continuity. Corollary 7.2 therefore implies that at any global maximum q,

$$(a \cdot \alpha)(q \cdot \alpha) \geq 0 \qquad \text{for every reflection normal } \alpha. \tag{7.2}$$

Otherwise reflection would strictly increase the value.

If $\alpha_i \cdot q < 0$, dominance of a and (7.2) force $\alpha_i \cdot a = 0$. Reflecting across this wall therefore fixes a and preserves the maximal value. The terminating simple-reflection procedure from Section 3 gives $w \in W_a$ with $wq \in \overline{C}$. By Theorem 6.3, $q \in \partial P$; since w preserves P, one has $wq \in \partial P \cap \overline{C} = \Delta$. Consequently, $q \in w^{-1}\Delta$ with $w^{-1} \in W_a$, proving the maximum-set containment. Compactness and continuity give nonemptiness. Central oddness of f_a gives the exact relation between the maximum and minimum sets.

For a strictly dominant parameter the stabilizer is trivial. For $a = \lambda b_3$, its stabilizer $\langle s_1, s_2 \rangle$ moves Δ through the ten flag triangles of the physical pentagonal face Γ. This proves the last assertions and Theorem 1.3. □

7.3. **Strict tangential inequalities on the physical face.** The physical triangle has a convenient pair of tangential coordinates. Set

$$n = b_3, \qquad t = e_y - \varphi e_z, \qquad T_0 = \frac{1}{\varphi(\varphi+2)}, \qquad h_0 = \frac{1}{2\varphi+1}.$$

Both e_x and t are orthogonal to n, and direct substitution in (1.6) gives

$$q = c + \xi e_x + \eta t \in \Delta \quad \Longleftrightarrow \quad 0 \leq \eta \leq T_0, \qquad 0 \leq \xi \leq \frac{h_0}{T_0}\eta. \tag{7.3}$$

In these coordinates $c = (0,0)$, $e = (0, T_0)$, and $v = (h_0, T_0)$. Thus $\Delta \setminus [e, v]$ lies in the relative interior of the full physical face Γ, including the two internal mirror sides and the face center.

Lemma 7.4 (Strictness of a tangential cone functional)**.** *If $L \in \mathcal{K}_D^*$, $\mathcal{Q}L = L$, and $\operatorname{tr} L > 0$, then*

$$\langle L, DF(q) \rangle > 0 \qquad (q \in \Delta \setminus [e, v]).$$

The derivative matrix at these points is that of the smooth even Neumann extension across the open physical face.

Proof. For any constant matrix H, the condition $\mathcal{Q}H = H$ is equivalent to $Hn = 0$. Every row of such a matrix is tangent to the physical face, so $g_H = \langle H, DF \rangle$ is a sum of tangential derivatives of the components of F. Their smooth even Neumann extensions across the open face show that g_H extends smoothly and evenly, solves $-\Delta g_H = \mu g_H$, and has homogeneous Neumann data there.

Fix $q \in \Delta \setminus [e, v]$ and put $J = \{i : \alpha_i \cdot q = 0\} \subset \{1, 2\}$. Use the stabilizer W_J and orthogonal projection P_J from Lemma 6.1, with $P_J = I$ when J is empty. That construction preserves the dual cone and trace. Moreover, s_1, s_2 fix n, so their conjugation projections commute with $\mathcal{Q}$. Their cyclic limit P_J commutes with it as well. Hence

$$L' = P_J L \in \mathcal{K}_D^*, \qquad \mathcal{Q}L' = L', \qquad \operatorname{tr} L' = \operatorname{tr} L > 0.$$

The same strictness lemma gives $g_{L'} > 0$ in T. Choose a ball about q avoiding every other physical face and every nonincident mirror. Its intersection with P is a half-ball. The local chamber decomposition from Section 3, continuity on the chamber closure, and W_J-invariance of $g_{L'}$ make this function nonnegative on the whole half-ball. It is nonzero there because the half-ball meets T. The preceding tangential extension argument gives its weak Neumann condition, so Lemma 7.1 yields $g_{L'}(q) > 0$. Finally, smooth reflection makes DF continuous up to this open physical face. Taking interior limits in (3.4) shows that $DF(q)$ is fixed by W_J. Orthogonality therefore gives

$$\langle L, DF(q)\rangle = \langle P_J L, DF(q)\rangle = g_{L'}(q) > 0.$$

This also treats points on internal sides and the point c. □

Corollary 7.5 (The needed face directions)**.** *On $\Delta \setminus [e, v]$ one has*

$$D_{e_x} f_{b_1} > 0, \qquad D_t f_{b_1} > 0, \qquad D_t f_{b_2} > 0. \tag{7.4}$$

Proof. The three relevant matrices have the exact decompositions

$$\begin{aligned} b_1 e_x^T &= L_5, \\ b_1 t^T &= (\varphi + 2) L_9, \\ b_2 t^T &= (3 - \varphi) L_6 + (4\varphi - 2) L_9. \end{aligned}$$

The middle identity follows from $\mathcal{Q}(b_1 e_y^T) = b_1 t^T/(\varphi + 2)$; the other two are part of the exact certificate. All coefficients are positive. The three matrices are fixed by $\mathcal{Q}$ and their traces are respectively 1, φ^2, and 2φ. Lemma 7.4 applies. □

7.4. **Tangential differentiation along a physical edge.**

Lemma 7.6 (Weak Neumann equation for an edge-tangential derivative)**.** *Let $u \in H^2(P)$ be a Neumann eigenfunction with eigenvalue $\mu \geq 0$, and let E be an open edge of the convex polyhedron P. Let t be a constant unit vector parallel to E. On every sufficiently small neighborhood of a point of E that avoids its endpoints, $v = D_t u \in H^1$ satisfies*

$$-\Delta v = \mu v$$

with homogeneous weak Neumann conditions on the two incident facets. The derivative v has a locally Hölder continuous representative up to the edge. On the edge, the restriction of u is classically differentiable in direction t, with derivative v. If v is also nonnegative and nonzero on such a connected neighborhood inside P, it is strictly positive at the edge.

Proof. In orthogonal coordinates whose last axis is t, the local domain is the product of a planar dihedral wedge with an interval. Choose a smaller product neighborhood whose closure avoids the artificial ends of that interval. Translations by sufficiently small multiples of t preserve the wedge and hence both physical facets. If η is an H^1 test function supported away from the artificial boundary of this smaller neighborhood, its translates are admissible local Neumann test functions. Subtracting the weak equations, with a change of variables in the translated equation, gives

$$\int \nabla \left(\frac{u(x + ht) - u(x)}{h} \right) \cdot \nabla \eta = \mu \int \frac{u(x + ht) - u(x)}{h} \, \eta.$$

The integrals are over the smaller neighborhood intersected with P. This includes all such test functions with nonzero traces on the two physical facets. As $h \to 0$, the difference quotients converge to $D_t u$ in H^1 on the smaller neighborhood. Indeed,

$$\frac{u(x+ht)-u(x)}{h} = \int_0^1 D_t u(x+\theta ht)\, d\theta,$$

and $u \in H^2$ gives translation continuity in L^2 for $D_t u$ and its first derivatives along this product neighborhood. Passing to the limit proves the asserted local weak Neumann equation.

Lemma 7.1 supplies a continuous representative of v at the edge without any sign assumption, and gives strict positivity under the additional sign hypothesis. No global C^1 regularity of u on the polyhedron is needed. For a compact subsegment $[p, p+\ell t]$ of the open edge, choose a fixed vector pointing into its transverse wedge and translate the segment a distance ε in that direction. The resulting parallel segments lie in P. Interior smoothness and the fundamental theorem of calculus give

$$u(p_\varepsilon + \ell t) - u(p_\varepsilon) = \int_0^\ell v(p_\varepsilon + st)\, ds.$$

The eigenfunction is continuous on $\overline{P}$, and v is continuous on a compact neighborhood of the subsegment relative to $\overline{P}$. Uniform continuity there allows $\varepsilon \downarrow 0$ in the integral, giving the same identity on the edge. Continuity of the integrand proves the claimed classical differentiability along the edge. □

Corollary 7.7 (Strict increase from the edge midpoint to the vertex)**.** *In the normalized flag triangle, the restriction of f_{b_1} to the segment $[e, v]$ is strictly increasing from e toward v.*

Proof. By the edge geometry in Section 3, the segment is parallel to e_x, which is tangent to both incident physical faces. The matrix $L_5 = b_1 e_x^T$ belongs to $\mathcal{K}_D^*$ and has trace 1. Lemma 6.1 therefore gives

$$D_{e_x} f_{b_1} > 0 \quad \text{in } T.$$

The reflection s_3 fixes b_1 and e_x, so this derivative is even across $z = 0$. At a point of (e, v), the incident mirror stabilizer is $\langle s_3 \rangle$. Its local chamber decomposition therefore makes the derivative nonnegative on a full neighborhood inside P, and it is nonzero there by its positivity in T. Lemma 7.6 shows that it remains strictly positive at the open edge. Integration gives strict increase between any two points of the relative interior. Continuity of f_{b_1} at e and v, together with a strictly intermediate open subsegment, gives strict increase also when one or both of the comparison points are endpoints. □

7.5. **Exact vertex extrema and the edge parameter sector.**

Theorem 7.8 (Unique extrema for every vertex-axis mode)**.** *For every $w \in W$ and $\lambda > 0$,*

$$\mathcal{M}(\lambda w b_1) = \{wv\}, \qquad \Updownarrow(\lambda w b_1) = \{-wv\}.$$

In particular f_{b_1} has its unique global maximum at v and its unique global minimum at $-v$.

Proof. First maximize f_{b_1} on Δ. For $q = c + \xi e_x + \eta t \in \Delta \setminus [e, v]$, the direction to v is

$$v - q = (h_0 - \xi) e_x + (T_0 - \eta) t.$$

Both coefficients are nonnegative, and the second is positive. The segment lies in Δ and, until its terminal point, in the relative interior of the physical pentagonal face. Corollary 7.5 gives strict increase of f_{b_1} along the segment before the endpoint. Continuity at the endpoint, and comparison with an intermediate point, give $f_{b_1}(q) < f_{b_1}(v)$. For $q \in [e, v)$, the same strict comparison follows from Corollary 7.7. Thus v is the unique maximizer on Δ.

Theorem 7.3 confines the global maxima to translates of Δ under $W_{b_1} = \langle s_2, s_3 \rangle$. This subgroup fixes v and preserves f_{b_1}. Every such translated triangle therefore has the same unique maximizer v.

This proves uniqueness on $\overline{P}$. Central oddness gives the unique minimum $-v$. Finally covariance and positive scaling give the statement for $\lambda w b_1$. □

Theorem 7.9 (Edge localization on a parameter mirror)**.** *Suppose $a = t_1 b_1 + t_2 b_2$, where $t_1, t_2 \geq 0$ and $a \neq 0$. If $t_1 > 0$, then*

$$\mathcal{M}(a) \subset (e, v]. \tag{7.5}$$

If $t_1 = 0$, then

$$\mathcal{M}(a) \subset E = [s_1 v, v]. \tag{7.6}$$

The corresponding minimum sets are the antipodes of the maximum sets. These are containing intervals, except for the exact vertex-axis case in the preceding theorem.

Proof. By Corollary 7.5 and linearity, $D_t f_a > 0$ on $\Delta \setminus [e, v]$. If $q = c + \xi e_x + \eta t$ is in that set, increase η with ξ fixed until it reaches T_0. The inequalities in (7.3) show that this segment stays in Δ, and its endpoint is on $[e, v]$. Strict increase before the endpoint and continuity give a larger value there. Thus any maximum on Δ lies on $[e, v]$.

If $t_1, t_2 > 0$, the stabilizer of a is $\langle s_3 \rangle$, which fixes $[e, v]$ pointwise. The general localization theorem therefore confines all global maxima to this half-edge. If $t_1 = 0 < t_2$, the stabilizer is $\langle s_1, s_3 \rangle$, and its translates of $[e, v]$ form the full physical edge E. If $t_2 = 0 < t_1$, the preceding theorem already gives the unique maximum v.

It remains to exclude e when $t_1, t_2 > 0$. Using $s_1 b_1 = b_1 - 2e_x$ and $s_1 e_x = -e_x$ gives

$$\mathcal{P}_1 L_5 = e_x e_x^T \in \mathcal{K}_D^*, \qquad \operatorname{tr}(e_x e_x^T) = 1.$$

Consequently $g = D_{e_x} f_{e_x}$ is strictly positive in T. This function is even under s_1 and s_3: f_{e_x} is odd in the x coordinate and even in the z coordinate, while differentiating in x reverses the former parity. The four local chambers associated with these two mirrors fill the local physical wedge around e. Thus $g \geq 0$ in a full neighborhood of e inside P and is nonzero there. The vector e_x is tangent to both incident physical faces. Lemma 7.6 gives

$$D_{e_x} f_{e_x}(e) > 0.$$

The same lemma, using its regularity assertion without a sign assumption, gives a continuous edge derivative of f_{e_y}. Since f_{e_y} is even under s_1, that derivative is zero at the edge midpoint e. As $a = (t_1, \varphi^2 t_1 + 2\varphi t_2, 0)$, linearity gives

$$D_{e_x} f_a(e) = t_1 D_{e_x} f_{e_x}(e) > 0.$$

Moving a short positive distance along the physical edge therefore increases the value, excluding e as a maximum. Finally central oddness gives the assertions about minima. Together with the preceding theorem this proves Theorem 1.4. □

7.6. **Further comparisons and the complete list of proved regions.**

Proposition 7.10 (Exclusion of internal sides on an open face)**.** *Let $a \in \overline{C} \setminus \{0\}$ and let q be in the relative interior of the physical face Γ. If $\alpha_i \cdot q = 0$ and $\alpha_i \cdot a > 0$ for either $i = 1$ or $i = 2$, then q cannot be a global maximum of f_a.*

Proof. For $i = 1, 2$, the vector α_i is tangent to Γ. Even reflection across that physical face extends f_{α_i} smoothly to a neighborhood of q. The reflection preserves the half-space $\alpha_i \cdot x > 0$. Corollary 7.2 and even reflection make the extended solution positive on that side and zero on the internal mirror. On a small half-ball bounded by the mirror it is nonnegative and satisfies $\Delta f_{\alpha_i} = -\mu f_{\alpha_i} \leq 0$. The Hopf boundary point lemma on this smooth half-ball yields

$$D_{\alpha_i} f_{\alpha_i}(q) > 0.$$

Write $a = (a \cdot \alpha_i)\alpha_i + a_\perp$. The function $f_{a_\perp}$ is even across the mirror; its smooth face extension therefore satisfies $D_{\alpha_i} f_{a_\perp}(q) = 0$. Hence

$$D_{\alpha_i} f_a(q) = (a \cdot \alpha_i) D_{\alpha_i} f_{\alpha_i}(q) > 0.$$

This is a direction tangent to the physical face at one of its relative interior points, so it excludes a maximum. □

Proposition 7.11 (Values at all vertices)**.** *There exists $\beta > 0$, independent of the vertex q, such that*

$$F(q) = \beta q, \qquad f_a(q) = \beta\, a \cdot q \qquad (q \text{ a vertex of } P). \tag{7.7}$$

Thus, among the vertices, the maximizing vertices are exactly those that maximize the linear functional $x \mapsto a \cdot x$.

Proof. The stabilizer of v is generated by s_2, s_3, so its fixed space on $\mathbb{R}^3$ is $\alpha_2^\perp \cap \alpha_3^\perp = \mathbb{R}v$. Covariance and transitivity on the vertices therefore imply $F(q) = \beta q$ with a common real coefficient β. At v the first root coordinate is positive, and Corollary 7.2 gives

$$0 < f_{\alpha_1}(v) = \alpha_1 \cdot F(v) = \beta\, \alpha_1 \cdot v.$$

Hence $\beta > 0$. Pairing with a proves the result. □

For an arbitrary nonzero parameter, first apply a symmetry to put it in the form

$$a = t_1 b_1 + t_2 b_2 + t_3 b_3, \qquad t_i = \alpha_i \cdot a \geq 0.$$

The stabilizer is generated by the simple mirrors for which $t_i = 0$. Table 3 lists all seven nonzero coefficient strata and combines Theorems 7.3, 7.8, and 7.9. The two-triangle regions in its second and third rows lie in the same full face because s_1 and s_2 both fix b_3. All regions for minima are the antipodes of the listed regions.

TABLE 3. Regions proved to contain every global maximum. The symbol + means strictly positive. Only the singleton row asserts equality with the full maximum set.

t_1	t_2	t_3	Region containing $\mathcal{M}(a)$
+	+	+	Δ
0	+	+	$\Delta \cup s_1\Delta$
+	0	+	$\Delta \cup s_2\Delta$
+	+	0	$(e, v]$
+	0	0	$\{v\}$, the unique maximum
0	+	0	$E = [s_1 v, v]$
0	0	+	Γ, the full pentagonal face

Within the relative interior of a physical face, the additional exclusion in Proposition 7.10 removes a nonphysical mirror side whenever its coefficient is positive. No exclusion at an actual physical edge or vertex is inferred from that proposition.

Remark 7.12 (Extent of the location results). The hot spots assertion is proved for the entire first eigenspace. The vertex-axis extrema are classified exactly. For other parameters the theorems locate all extrema in the stated regions, without specifying every maximizing point within them. In particular, the edge-axis maximum is not identified here with the edge midpoint, nor the face-axis maximum with the face center. Formula (7.7) compares vertex values; it does not imply that every global maximum must be a vertex. A complete pointwise classification for arbitrary parameters remains beyond the conclusions of this paper.

7.7. **Illustrations of the proved locations.** The following TikZ drawings are geometric projections of the regular dodecahedron and its pentagonal faces. Red marks maxima and blue marks minima. Figure 2 shows exact unique extremal points. In the other figures, a colored face or edge is a set proved to contain all corresponding extrema, and its shading does not assert that each of its points is extremal. Figure coordinates are used only for illustration; the proofs use the exact coordinates above.

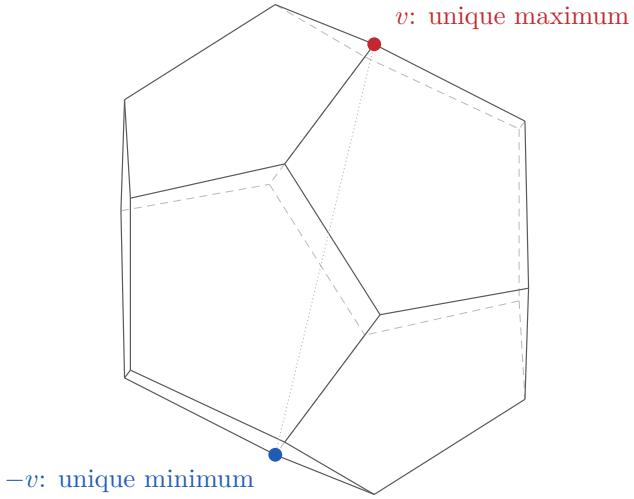


FIGURE 2. Exact extremal positions for a positive vertex-axis mode. The red vertex is the unique global maximum and its antipodal blue vertex is the unique global minimum. The drawing is an orthographic projection of the regular dodecahedron; rear edges are dashed.

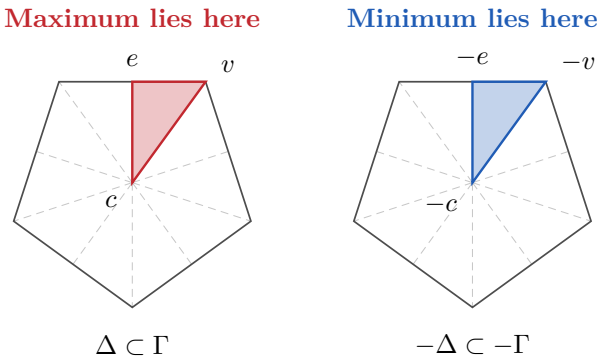
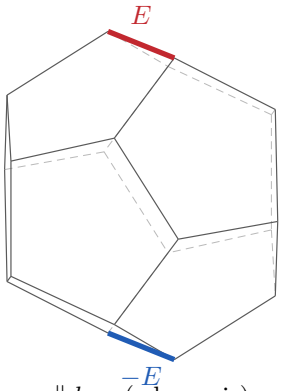
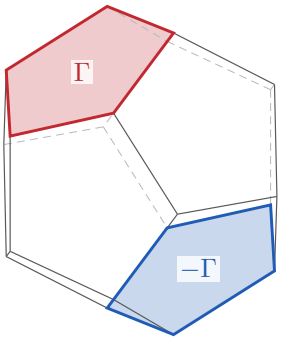


FIGURE 3. Localization for a strictly dominant coefficient vector $a \in C$. The red flag triangle contains every maximum; the antipodal blue triangle contains every minimum. The two pentagonal faces are shown in corresponding local coordinates. Shading denotes a containing region, not a set all of whose points are extrema. The other dashed subdivisions show the ten flag triangles of a face.



$a \parallel b_2$ (edge axis)
Maxima in E; minima in $-E$



$a \parallel b_3$ (face axis)
Maxima in Γ; minima in $-\Gamma$

FIGURE 4. Containing regions for positive edge-axis and face-axis modes. Here $E = [s_1 v, v]$ is one physical edge and $\Gamma = \{x \in \overline{P} : b_3 \cdot x = 1\}$ is one physical pentagonal face. Red contains every maximum and blue contains every minimum. The statements do not identify the precise maximizing point along E or within Γ. Rear highlighted regions are displayed through the wireframe for visibility.

REFERENCES

[1] M. S. Ashbaugh, F. Gesztesy, M. Mitrea, and G. Teschl, *Spectral theory for perturbed Krein Laplacians in nonsmooth domains*, Adv. Math. **223** (2010), 1372–1467. https://arxiv.org/abs/0907.1442.

[2] D. Gilbarg and N. S. Trudinger, *Elliptic Partial Differential Equations of Second Order*, Classics in Mathematics, Springer-Verlag, Berlin, 2001. https://doi.org/10.1007/978-3-642-61798-0.

[3] J. E. Humphreys, *Reflection Groups and Coxeter Groups*, Cambridge Studies in Advanced Mathematics, vol. 29, Cambridge University Press, Cambridge, 1990. https://doi.org/10.1017/CBO9780511623646.

[4] J. B. Kennedy and J. Rohleder, *On the hot spots conjecture in higher dimensions*, arXiv:2410.00816v2, 2025. https://arxiv.org/abs/2410.00816.